

Title: Synthesis of a Diagnostic Quality 12-Lead ECG Using a Generative Hybrid Convolutional and Long Short-Term Memory Neural Network

Authors: Tanay Chitlur, Ajay Tripuraneni, MD FHRS

Cardiovascular disease (CVD) is the leading cause of death in the United States. However, early detection can prevent up to 80% of associated fatalities. 12-lead Electrocardiograms (ECGs) are currently the gold standard for CVD diagnosis but require special equipment to be recorded. Although smart wearable devices are one of the best methods for continuous monitoring, they can only record a few leads (e.g., Apple Watch, Kardia), limiting the diseases that can be detected.

We propose a comprehensive method to synthesize a diagnostic quality 12-lead ECG from a subset of leads. We used three recordable leads, I, III, and V6, to generate the remaining nine leads. Leads II, aVR, aVL, and aVF are computed using Einthoven and Goldberger Formulas. Leads V1-V5 are synthesized using a novel machine-learning pipeline, the primary focus of our study.

The model implements a hybrid Convolutional (CNN) and Long-Short Term Memory (LSTM) Neural Network model, using individual biometrics and innovative feature extraction techniques, to capture spatial and temporal ECG features, resulting in a personalized, diagnostic quality ECG output. The model is trained using ~20,000 ECGs from PTB-XL database.

The hybrid CNN + LSTM model employs a cross-correlation architecture, extracting individual lead features and cross-correlating them with other leads. The hybrid architecture forms intricate connections within the model, creating a more accurate synthesized ECG.

The model achieved a low error with an average Normalized Root Mean Squared Error of 0.106. The synthesized ECG also captures the morphology of both normal and abnormal ECGs. These results demonstrate that the innovative architecture is able to generate accurate 12-lead ECGs but to find the best method other generative models (e.g. transformers) must be tested. The impact of the proposed generative AI methodology has immense making early detection and continuous monitoring of CVD feasible via wearable devices.

References:

Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., ... & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals.